

EP 302: Computational Methods for Engineering

Course Code	EP 302
Course Name	Computational Methods for Engineering
L-T-P-C	2-0-2-3
Prerequisites	IC 110 Engineering Mathematics
Intended for	UG
Distribution	Core
Approval	4th BoA (Feb 2015)

Course Contents

- **Module I**
Numerical techniques: Linear and non-linear equations; Solving linear systems: Gauss Elimination, Gauss Jordan, LU Decomposition; Solving linear equations with iterative methods: Jacobi method, Gauss-Seidal method, Successive Over Relaxation (SOR); Finding roots of polynomial and transcendental equations: Bisection method and Newton-Raphson Method; Numerical Integration: Trapezoidal rule, Simpson's rule, Gaussian Quadrature; Case study from multi-physics systems. [14 Lectures]
- **Module II**
FEM Concept: Introduction, Engineering applications of finite element method, Weak formulation, Interpolation scheme, FEM formulation for 1D and 2D problems, Computer implementation issues, Convergence and Error analysis. [6 Lectures]
- **Module III**
FEM for Multi-Physics Systems: Case study of thermo-mechanical systems, electromechanical systems, thermo-electro-mechanical systems. [5 Lectures]
- **Module IV**
Simulation of Engineering Systems: Monte-Carlo simulation, Simulation of continuous and discrete processes with suitable examples from engineering problems. [3 Lectures]

Laboratory sessions

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- Exp. 1 & 2 Use of numerical techniques to solve system of equations using computer programming platform.
- Exp. 3 & 4 Introductions to computational software (FEM based packages).
- Exp. 5, 6 & 7 Find the simulation response of a physical system under (i) Mechanical load (ii) Thermal load (iii) Electrical load environment.
- Exp. 8 & 9 Use of computational (FEM based) package to simulate multi-physics systems case e.g. Electric motor under electro-mechanical and thermal environment.
- Exp. 10 & 11 Monte Carlo simulation.

Text Books

1. JN Reddy, **Introduction to Finite Element Methods**, 3rd Edition, McGraw Hill, 2005.
2. Steven C. Chapra and Raymond P. Canale, **Numerical methods for Engineers**, 4th Edition, Wiley 2015.

References

1. Conte and de Boor, **Elementary Numerical Analysis: Algorithmic Approach**, 3rd Edition, McGraw Hill, 1980.
2. Rubinstein, **Simulation and the Monte Carlo method**, 2nd Edition, Wiley 2007.
3. Smith, Griffiths and Margetts, **Programming the Finite Element Method**, 5th Edition, Wiley, 2013.
4. Anju Khandelwal, **Numerical Methods and Computer Programming**, 4th Edition, Alpha Science International Ltd. 2015.